



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
HALF YEARLY EXAMINATION

CLASS: X
SUBJECT: BIOLOGY [SET A]

FULL MARKS: 80
TIME: 2 HOURS

*Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent reading the Question Paper.
The time given at the head of this Paper is the time allowed for writing the answers.*

*Section A is compulsory. Attempt any four questions from Section B.
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of seven printed pages.*

SECTION A

(Attempt all questions from this Section)

Question 1.

Choose the correct answer to the questions from the given options. (Do not copy the question, write the correct answer only):

[15]

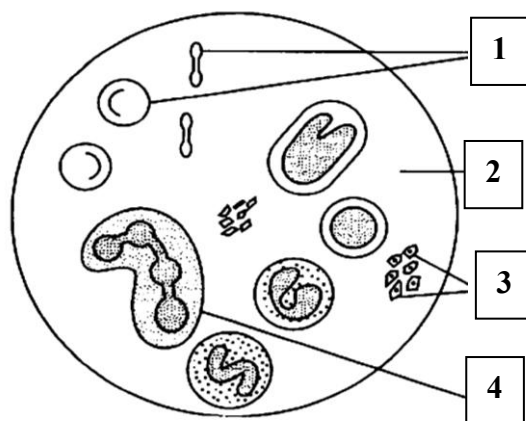
- (i) A cell has eight pairs of chromosomes. After meiosis, the number of chromosomes in each daughter cells will be:
(a) Five (b) Four (c) Sixteen (d) Eight
- (ii) Number of water molecules required to make one molecule of glucose during photosynthesis:
(a) 12 (b) 6 (c) 1 (d) 0
- (iii) Akash tried to understand the function of two glands in male body. He tabulated the pairs as follows:

Gland	Function
P	Secretion acts as a medium to transport sperm
Q	Neutralizes the acidic condition of vagina

- (a) P-Seminal vesicle and Q- Bulbo Urethral gland
(b) P- Cowper's gland and Q- Prostate gland
(c) P- Prostate gland and Q- Seminal vesicle
(d) P- Seminal vesicle and Q- Prostate gland

- (iv) Ravi had done his blood test, and the report suggested that there was an abnormal increase in the number of RBCs in his blood. He went to a doctor and found out he was suffering from _____. Identify the type of blood abnormality detected by the doctor.
- (a) Polycythaemia (b) Erythropenia
(c) Haemopenia (d) Leukopenia
- (v) Assertion (A): Glucagon is secreted by alpha cells of Pancreas to raise glucose level in the blood.
Reason (R): Glucagon breaks down glycogen to glucose in the liver.
- (a) A is true and R is false (b) Both A and R are true
(c) A is false and R is true (d) Both A and R are false
- (vi) Rohan had done a cross between a pure tall plant and a pure dwarf plant. He and his friends tried to study the phenotype of the offspring formed by this cross. Identify the person who gave the correct Phenotype of F₁ generation?
1. Rina- 3 Tall: 1 Dwarf 2. Rohan- All Homozygous dwarfs
3. Sohini- All heterozygous tall 4. Alam- One tall and three dwarfs
- (a) Rina is correct (b) Alam is correct
(c) Sohini is correct (d) Rohan is correct
- (vii) If a pure round seed (RR) plant is crossed with a pure wrinkled seed plant (rr), the phenotypic ratio (P) and genotypic ratio (Q) of F₂ generation will be?
- (a) P- All Round seed and Q- All Rr
(b) P- 1 Round seed: 3 Wrinkled seed and Q- 1RR: 2Rr: 1rr
(c) P- 3 Round:1 Wrinkled and Q- 1 RR:2 Rr: 1rr
(d) P- 2 Round: 1 Wrinkled and Q- 2 RR: 1rr
- (viii) Among the following which nitrogenous bases belong to the group Pyrimidine?
P. Uracil Q. Adenine R. Thymine S. Cytosine
- (a) Both P and Q (b) P, R and S
(c) Both Q and S (d) Q, R and S
- (ix) Assertion (A)- Imbibition is a passive absorption of water by substances such as cellulose present in plant.
Reason (R) - Cellulose is a hydrophilic substance.
- (a) A is true and R is false (b) Both A and R are true
(c) A is false and R is true (d) Both A and R are false
- (x) Vijay sells mangoes in the market. He was sad that the mangoes of the trees were not ripening and thus, he could not sell the mangoes. His friend suggested to use a plant hormone which would help the mangoes to ripen faster. Identify which hormone he had used to ripen the mangoes.
- (a) Ethylene (b) Cytokinin
(c) Absciscic acid (d) Gibberellins

- (xi) The compound formed when oxygen combines with haemoglobin is P and the compound formed when carbon monoxide combines with haemoglobin is Q
 (a) P- Carbaminohaemoglobin and Q- Oxyhaemoglobin
 (b) P- Oxyhaemoglobin and Q- Carboxyhaemoglobin
 (c) P-Oxyhaemoglobin and Q- Carbaminohaemoglobin
 (d) P- Carbohaemoglobin and Q- Oxyhaemoglobin
- (xii) The hormone which helps in formation of egg in females:
 (a) TSH (b) Glucagon
 (c) Oxytocin (d) FSH
- (xiii) A normal human male has P autosomes and Q allosomes
 (a) P- 44 and Q - XX (b) P- 23 and Q - XY
 (c) P- 23 and Q- XX (d) P- 44 and Q- XY
- (xiv) Identify from the following diagram the initiator of blood clotting process.



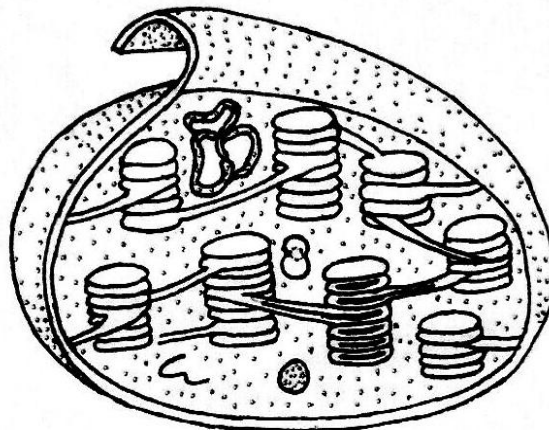
- (a) Only 1 (b) Only 3
 (c) Only 2 (d) Only 4
- (xv) Assertion (A) – When we swallow the food, the digested food from mouth is pushed through the oesophagus by peristalsis movement.
 Reason (R) – Food moves down through the oesophagus due to Natural reflexes.
 (a) A is true and R is false (b) A is false and R is true
 (c) Both A and R is true (d) Both A and R are false.

Question 2

- (i) Give the biological/technical terms for the following: [5]
- (a) A soluble protein found in the blood plasma.
 (b) A sheet of fibres connecting two cerebral hemispheres.
 (c) A condition in which blood passes through urine due to infection.
 (d) The process through which WBC squeezes through the walls of the capillaries.
 (e) The part of the chloroplast where the light independent reaction of photosynthesis takes place.

- (ii) Given below is the diagram of a chloroplast. Read the information given below the diagram and fill in the blanks:

[5]



The formation of ATP from ADP is known as __ (a) __. The end product of photosynthesis is glucose, which is converted to starch by the process known as __ (b) __. To test for the presence of starch we use __ (c) __ solution which is brown in colour and in the presence of starch this solution turns __ (d) __ in colour. Full form of NADP is __ (e) __.

- (iii) Choose the odd one out and write the correct category to which the others belong: [5]

- (a) Pons, Corpus luteum, Cerebrum, Cerebellum.
- (b) Vas deferens, Fallopian tube, Epididymis, Cowper's gland.
- (c) GH, TSH, Vasopressin, LH.
- (d) Incus, Malleus, Pinna, Stapes.
- (e) Eosinophils, Monocytes, Basophils, Neutrophils.

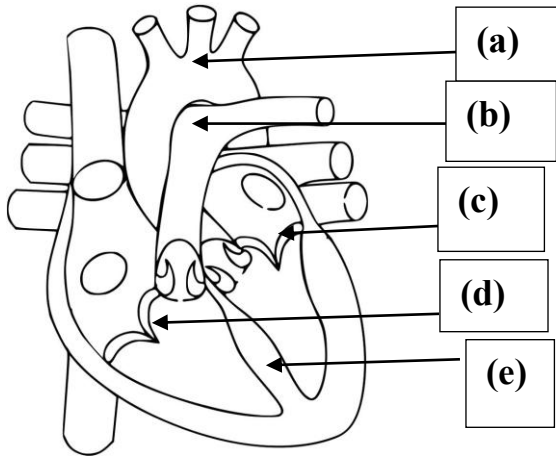
- (iv) In the biology class 10 test, there were few questions on the structure of the eye. The teacher wrote the questions on the board and asked the students to solve them. Ishayu was not being able to solve the questions. Help him in solving the following questions:

[5]

- (a) Name the enzyme present in the tear.
- (b) Name the nerve which carries information from eye to the brain.
- (c) Which layer of the eye is sensitive to light?
- (d) Name the pigment which is found in cones.
- (e) Name the layer which is richly supplied with blood vessels.

- (v) Given below is the cross section of human heart. Match the structures marked (a) To (e) with their correct functions:

[5]

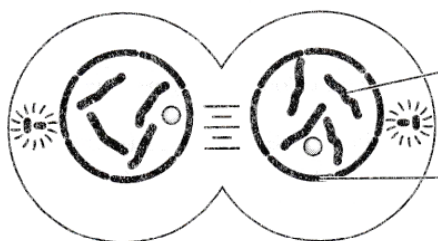
Cross section of Human Heart	Functions
	1. Separates the oxygenated and deoxygenated blood. 2. Only allows unidirectional flow of blood from right atrium to right ventricle. 3. Carries deoxygenated blood from all parts of the body to right atrium. 4. Only allows unidirectional flow of blood from left atrium to left ventricle. 5. Carries oxygenated blood to different parts of our body. 6. Carries deoxygenated blood to the lungs.

SECTION B

(Attempt any four questions from this section)

Question 3

- Mention the name of a non-circulating fluid present in the cavities of skeletal joints. [1]
- “Carbon monoxide is harmful for our body” Explain why. [2]
- A carrier woman (XX^0) and a colour blind man (X^0Y) produces offsprings. Find out the percentage of offsprings who are colour blind male and colour blind female. [2]
- Daraknath is a person from a hilly area. When a group of researchers went at his home, they saw Daraknath’s mother was consuming iodised salt in her diet. They said that people in hilly areas take iodine in their diet. Explain the statement with two reasons. [2]
- Answer the following questions by observing the diagram: [3]



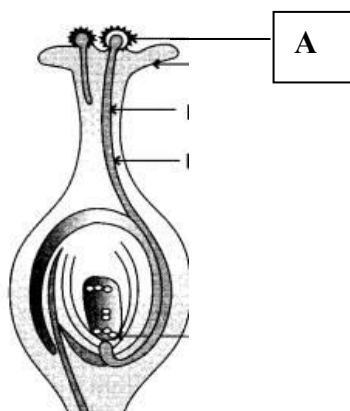
- Draw the diagram of the phase which comes prior to this phase.
- Mention one main property of the above phase shown in the diagram.

Question 4

- (i) Define genotype. [1]
- (ii) Tropic hormones are important for proper functioning of our body. Write the full forms of FSH and TSH. [2]
- (iii) Brain is an important organ of our body. It controls all the activities of our body. Students of Class X were doing research on the presence of white matter and gray matter in their brain. Help the students of class X in understanding the location of white matter and gray matter in the brain. [2]
- (iv) "Mature Red Blood cells lack mitochondria". Give reasons to justify the statement. [2]
- (v) Draw a neat, labelled diagram of L.S. of kidney. [3]

Question 5

- (i) What do you understand by the term parturition? [1]
- (ii) Class X students had gone to Biology lab for their Starch test activity. There they had used methylated spirit during starch test. Why do we boil a leaf in methylated spirit during starch test? How do we destarch a plant? [2]
- (iii) Explain the Law of Dominance. [2]
- (iv) Genetics is a branch of science which deals with the study of genes. Name the Scientist who had discovered the basic principles of genetics for the first time. State the phenotypic ratio of a dihybrid cross in F₂ generation. [2]
- (v) Study the diagram given below and answer the following questions: [3]



- (a) Mention the tropism observed in the diagram above.
- (b) Define Tropic movement.
- (c) Give an example of a stimulant that helps in the growth of the part marked as 'A'.

Question 6

- (i) State the scientific name of the plant used by Mendel for his experiment. [1]
- (ii) Growth hormone is very important as it helps in the growth of a person. Mention the other name of this hormone and write the name of the gland which produces Growth Hormone. [2]
- (iii) Tear glands are present in the upper sideward portions of the orbit. It produces tears, which is also useful to us in many ways. Mention two important functions of tear. [2]
- (iv) Draw a flow chart to show the main reactions of blood clotting. [2]

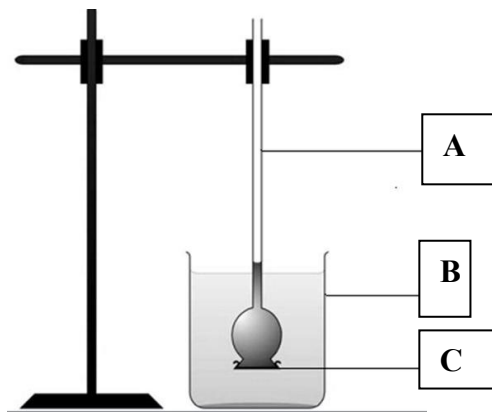
- (v) Sohan and his group of friends had crossed a pure plant with Axillary flower (A) with pure plant with Terminal flower (a) for their biology project. Help them in understanding the experiment by answering the following questions: [3]
- Mention the phenotypic and genotypic ratio of F_2 generation.
 - Mention the probability of gametes that are formed in the F_2 generation.

Question 7

- Why are the leaves of *Geranium* plant known as variegated leaves? [1]
- Virat was not able to see anything written on the board from the seventh row of his classroom, so his teacher asked his parents to take him to an ophthalmologist. The doctor told his parents that he has an eye defect. Mention the reasons for the cause of this eye defect. [2]
- Blood group "AB" is known as Universal recipient. Explain with reasons. [2]
- Define Transpiration. Why is the pot in the experiment of transpiration covered with plastic? [2]
- Draw a neat, labelled diagram of stomata. [3]

Question 8

- Mention the effect of high wind velocity on the rate of transpiration. [1]
- Observe the diagram given below and identify 'C' and give one example of "C". [2]



- Give reasons: [2]
 - Leaves of some plants are wavy and folded.
 - Dry cobalt chloride paper changes to pink colour faster on the lower surface of the leaves.
- Chromatin fibres are present inside the nucleus. They condense to form chromosome during cell division. What do you think of these chromatin fibres are made up of? [2]
- Answer the following questions: [3]
 - Too much fertilizer in the agricultural field kills crops.
 - Mention the structure present in the inner ear which helps in maintaining the dynamic balance of our body.